AI Procurement Request Copilot

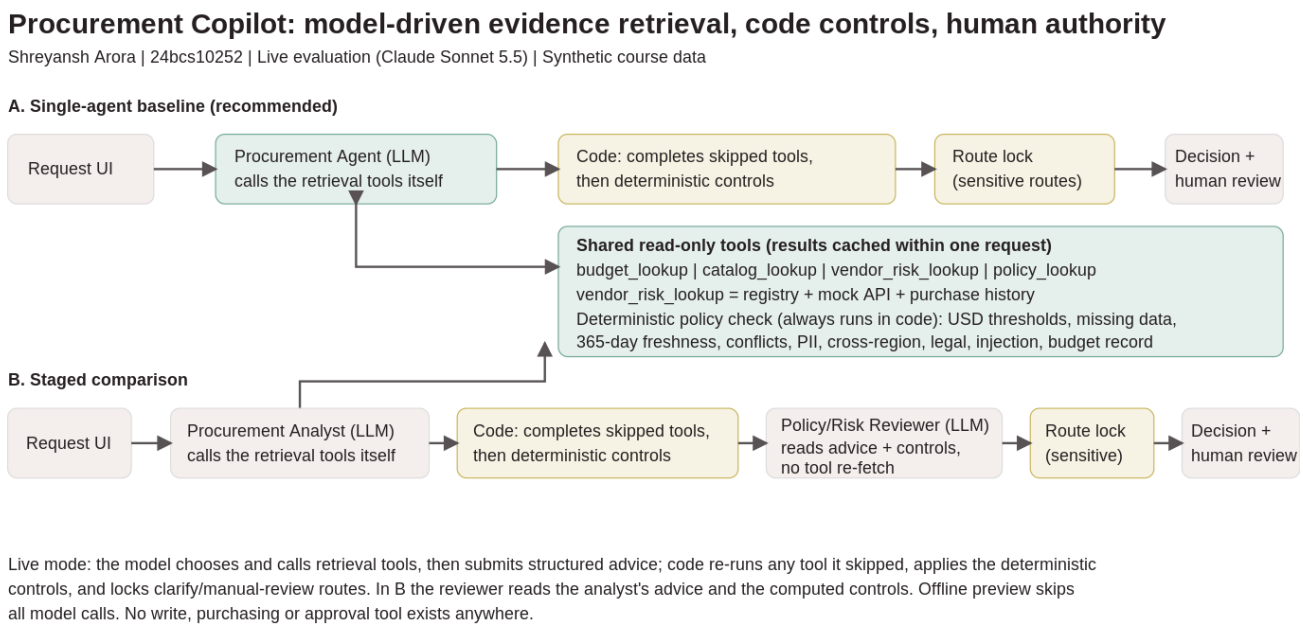
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Demo video: https://www.loom.com/share/55a6148d0bcc4e39b80ed5cffa3e2123

Problem and boundary

Companies receive ad-hoc software purchase requests and must route them through budget, catalog, vendor-risk and policy checks before a human decides. This copilot gathers that evidence and drafts a structured, schema-checked recommendation. Its boundary is hard: it advises and never purchases, approves spending, changes budgets, accepts terms or messages vendors. There is no approve button or purchasing tool; every recommendation ends in human review.

Architecture



The model (Claude Sonnet 5.5) retrieves evidence through four read-only tools - budget, catalog, vendor risk and policy - then submits schema-checked advice. Code re-runs any tool the model skipped, applies the deterministic controls (financial thresholds, evidence freshness, conflicting records, missing budget records, security/privacy/legal triggers, prompt-injection signal) and locks clarify and manual-review routes so model text cannot remove them. Architecture A is a single agent; B adds a Policy/Risk Reviewer over the same evidence pack without re-fetching tools.

Evaluation

Live run, Claude Sonnet 5.5, 72/72 valid decisions: six public cases plus four disclosed extensions, three repeats per architecture. 182 physical API calls (all HTTP 200). Reported usage 641,735 input / 61,489 output tokens, a $1.8984 estimate at $2/$10 per million tokens - not an invoice. Raw calls, ledger and per-run decisions are preserved under validation/anthropic_toolcalling/; 125 offline tests pass.

	A (single)	B (staged)
Final route = expected	30/30	30/30
Model route = expected	27/30	27/30
Controls preserved	30/30	30/30
Model calls	60	90
Median latency	4.92s	8.63s
p95 latency	6.38s	11.20s

An earlier code-orchestrated run (preserved in validation/anthropic_fresh/) reached the same final routes but the model never called tools, and its own route matched expectations in only 18/30 runs per architecture. Those corrections traced to that run's routing rules: its prompt only sent stale, conflicting or unavailable evidence to manual review, so the model largely followed its instructions. Separately, the wave-3 run exposed a missing precedence between the clarify and manual-review triggers; an explicit precedence line (clarify over manual review) has been added and is re-measured in the live verification reruns reported in the README. This table's wave-3 evidence predates the wave-4/5 fixes; the deterministic controls and final routes are unchanged by them.

Ship decision

Ship A with the route lock kept mandatory, as a cost/latency trade-off with a human-review backstop. Above, B used 90 model calls against A's 60 and 1.75x the median latency (8.63s vs 4.92s) with identical final routes; its reviewer changed the analyst's route in 0 of 36 runs. In wave 5 the reviewer changed it in 4 of 14 cases, and on four benign requests stating no business purpose B's stricter routes were correct by policy where A approved (corrected labels: A 10/14, B 14/14) - B's measured benefit. A business-purpose prompt rule was then written for those five benign cases (in-sample origin, disclosed; wave 5b: A 5/5, B 4/5). Wave 7 re-verified the final prompt on the comparison set, one repeat (50 calls, $0.5629 est.): final routes matched the archived run on A 10/10 and B 10/10 cases, expected on A 10/10 and B 10/10. Clarify and manual-review routes stay code-locked and every route ends with a human reviewer.

Limits

Synthetic snapshot data; ten requests; one provider and model; manual, non-independent semantic review; latency measured in one container against the mock vendor API. Missing for production: authorization, audit storage, real seat-use data, and broader adversarial and paraphrased tests. The injection regex was developed on the wave-5 phrasings (its development set); on an independent review's unseen sets it caught 3/12 and 0/15 attacks with 1/14 and 1/15 benign false positives - good precision, low recall. The model's own injection flag, a different detector, fired on 18/18 wave-5 attack decisions and 0/10 benign. Human reviewers still decide fit and exceptions.